

MALEK MOHAMMAD

Senior RAN Optimization Engineer | Telecommunications

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PROFESSIONAL SUMMARY

Results-driven telecommunications engineer with 3+ years of experience optimizing multi-technology RAN environments (2G/3G/4G/5G) at Zain Jordan. Proven track record in leading large-scale network transformation projects — including full vendor swap, 5G NSA deployment, and UMTS spectrum refarming — while concurrently developing internal automation platforms that significantly reduced manual workload. Combines deep radio expertise with strong software engineering skills to deliver data-driven performance improvements.

PROFESSIONAL EXPERIENCE

Senior RAN Optimization Engineer • Current

Zain Jordan | 2025 – Present

- Lead 5G NSA integration with Nokia, deploying the Thin-Layer concept to accelerate 100 MHz 5G expansion across the network.
- Execute LTE carrier load balancing strategies to optimize traffic distribution and improve user experience across key markets.
- Drive UMTS spectrum refarming initiatives to support UTRAN deployment and maximize spectrum efficiency.
- Manage RNC rehomings projects, reducing operational costs while maintaining network stability.

RAN Optimization Engineer

Zain Jordan | 2022 – 2025

- Owned optimization of the highest-traffic and VIP coverage areas in Zain Jordan's RAN, directly impacting premium customer experience.
- Monitored and benchmarked daily KPIs across 2G/3G/4G/NR to proactively detect and resolve performance degradation.
- Conducted drive testing campaigns using Nemo, TEMS, Spectrum Analyzers, and probes; translated field findings into actionable parameter optimizations.
- Designed and built NexusCore — a full-stack RNO automation platform — reducing engineer manual effort on repetitive CM workflows:
 - Frontend: JavaScript / CSS / HTML
 - Backend: Python + PostgreSQL, containerized on Docker
 - Authentication: LDAP integration with Zain employee directory (AAA)
- Built automated daily PM monitoring dashboards via SFTP pipelines and Power BI, covering all RATs across multiple network levels.
- Developed Nokia Configuration Manager — a custom UI tool for Nokia CM modules that improved RFE efficiency and parameter auditability.
- Delivered major network projects:
 - Full vendor swap: Huawei → Nokia
 - 5G FR1 rollout & L2100/L1800 new band deployments
 - GSM network deactivation & UMTS band re-farming

- RNC rehomings across multiple regions

PMO Engineer

Al Taka Electromechanical — Dubai, UAE | Feb 2016 – Sep 2017

- Coordinated integration of MEP infrastructure projects across multiple City of Dubai construction sites.

ACADEMIC & RESEARCH WORK

R&D Antenna Researcher

University of Jordan — EE Department | 2017 – 2022

- Designed a high-efficiency diversity antenna with configurable radiation patterns based on active feeding ports; extended to a 4x4 MIMO configuration with a fabricated prototype.
- Published research in three IEEE venues:
- Multi-Band Cylindrical Dielectric Resonator Antenna — IEEE AP-S/URSI 2022
- Multi-Port Multi-Band Pattern Reconfigurable Resonator Antenna — Cairo 2022
- Electrically Controlled Dual-Band Pattern Reconfigurable Antenna — IEEE MENACOMM 2022

EDUCATION

B.Sc. in Electrical Engineering

The University of Jordan | 2017 – 2022 | GPA: 3.2 / 4.0

CERTIFICATIONS

Nokia 5G Radio Planning — Advanced+

Nokia | Oct 2024

ID: dcb85c5d4f3bf35b35e3c81214d8c14a

Nokia 5G Radio Planning — Advanced

Nokia | Oct 2024

ID: e1a5ebaeed7e27e3a08f36dd00d06394

Nokia 5G Technology — Advanced

Nokia | Aug 2024

ID: cf86dda72ed08972b72fc4d014b83566

TECHNICAL SKILLS

RAN Tools & Platforms

- Nokia NetAct
- Huawei U2020
- Huawei PRS
- Astellia
- NEMO Analyze / Outdoor / Handy
- TEMS Discovery / Investigation
- ATOLL
- ANSYS HFSS

Software & Automation

- Python (advanced)
- SQL / PostgreSQL / SQLite
- Power BI / Tableau
- Docker
- JavaScript / HTML / CSS
- N8N Workflow Automation
- MATLAB
- Jupyter Notebook / VS Code

LANGUAGES

- Arabic — Native / Bilingual
- English — Full Professional Proficiency

VOLUNTEERING — IEEE UNIVERSITY OF JORDAN STUDENT CHAPTER

- IEEE PES Chapter Co-VP (2020–2021): Led field trips, organized events, and directed chapter annual planning.
- Event Coordinator (2018–2022): Managed workshops, field trips, and team selection for the IEEE UJSC PES chapter.
- Creative Graphic Designer (2018–2022): Produced the IEEE UJSB Spectrum monthly magazine using Adobe Illustrator.

REFERENCES

Available upon request. Contact details provided below:

- Eng. Mohammad Al-Kasem — RNO Senior Manager, Zain Jordan | mohammed.al-kasem@jo.zain.com
- Eng. Abdallah Zakarneh — RNO Division Team Lead, Zain Jordan | abdallah.zakarneh@jo.zain.com

- Eng. Thabet Jaradat — Technical Team Lead, Zain Jordan | thabet.jaradat@jo.zain.com
- Dr. Yanal Al-Faouri — Supervising Professor, University of Jordan | y.faouri@ju.edu.jo